PROJECT

Abstract

The main theorem in [Nak95] shows there is an action of the infinite-dimensional Heisenberg algebra on the homology groups of the Hilbert scheme of points on a projective surface. We explain how this representation is constructed and give an idea of the proof of the theorem.

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1. Introduction

This work is mostly based in Nakajima's book [Nak]. I have

2. Heisenberg algebra

In this section I review four different variants of the Heisenberg algebra and discuss their characters. We conclude showing how the character is used to prove that Nakajima's representation is in fact of highest weight.

The most basic definition of the Heisenberg Lie algebra is $\hat{\mathfrak{a}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n \oplus \mathbb{C}K$ with bracket given by

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0. \tag{2.0.1}$$

In class we described it's highest weight representation $H = \mathbb{C}[x_1, x_2, \ldots]$ with

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad K \mapsto \mathrm{Id}.$$

We introduced a grading in $H = \bigoplus_{\Delta=0}^{\infty} H_\Delta$ given by the *energy* of a monomial, defined by

$$\Delta(x_1^{k_1} x_2^{k_2} \ldots x_s^{k_s}) = \sum_j k_j,$$

which gives the character formula

$$\sum_{\Delta=0}^{\infty} q^{\Delta} \dim H_{\Delta} = \prod_{m=1}^{\infty} \frac{1}{1-q^m}. \tag{2.0.2}$$

In order to generalize this construction it's worth going over why this formula holds. As in the finite-dimensional case (see [Kac10, Lecture 24], Video 8), the highest weight representation of the Heisenberg algebra can be expressed as the universal enveloping algebra of the negative roots of the triangular decomposition. by the PBW theorem Writing $x_m = at^{-m}$ we have

$$\begin{aligned} H &\cong \mathbb{C}[at^{-1}, at^{-2}, \ldots] \\ &\cong \operatorname{Sym}\left(\bigoplus_{m>0} \mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0} \operatorname{Sym}(\mathbb{C}at^{-m}) \\ &\cong \bigotimes_{m>0} (\mathbb{C} \oplus \mathbb{C}at^{-m} \oplus \mathbb{C}(at^{-m})^2 \oplus \ldots), \end{aligned} \tag{2.0.3}$$

since the symmetric power of a vector space is the polynomial ring in its generators, and the symmetric power of a direct sum is the tensor product of the symmetric powers of the factors.

It is not a coincidence that we can write the Fock module of this algebra as a symmetric power of a Lie subalgebra. This is due to the Poincaré-Birkoff-Witten theorem applied to the universal enveloping algebra of the $\mathfrak{n}_-$ term of the triangular decomposition of $\hat{\mathfrak{a}}$ as in the theory of finite dimensional Lie algebras (see [Kac10, Lecture 24]).

Putting a dummy variably q to every 1-dimensional factor and multiplying throughout, we obtain

$$\begin{aligned} m &= 1 \qquad (1 + q + q^2 + q^3 + \ldots)\times \\ m &= 2 \qquad (1 + q^2 + q^4 + q^6 + \ldots)\times \\ m &= 3 \qquad (1 + q^3 + q^6 + q^9 + \ldots) \times \ldots, \end{aligned}$$

so that indeed the character is given by Equation 2.0.2.

Another version of this construction we discussed in lectures (see [vE23, Section 3.1]) starts with any finite-dimensional vector space $\mathfrak{h}$ equipped with a symmetric bilinear form $(\cdot,\cdot)$ and define

$$\hat{\mathfrak{h}} = \mathfrak{h} \otimes \mathbb{C}[[t^{\pm 1}]] \oplus \mathbb{C}K$$

with bracket

$$[at^m, bt^n] = m\delta_{m,-n}(a,b)K, \qquad [K, \hat{\mathfrak{h}}] = 0, \tag{2.0.4}$$

which is just the affine Lie algebra of $\mathfrak{h}$ when regarded as a commutative Lie algebra.

A similar formula to Equation 2.0.3 is

$$
\begin{aligned}
H &\cong \operatorname{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-n}\right)\\
&\cong \bigotimes_{m>0}\bigotimes_{a\in B}\operatorname{Sym}(\mathbb{C}at^{-m}) \qquad (2.0.5)\\
&\cong \bigotimes_{m>0}\bigotimes_{a\in B}\mathbb{C}[at^{-m}]
\end{aligned}
$$

where B is a basis of the underlying vector space $\mathfrak{h}$. If $\mathfrak{h}$ has dimension r, this has character

$$\prod_{m=0}^{\infty}\frac{1}{(1-q^m)^r}. \tag{2.0.6}$$

Taking this construction one step further, let's change $\mathfrak{h}$ for a finite-dimensional super vector space $\mathfrak{h}=\mathfrak{h}_0\oplus\mathfrak{h}_1$ with basis $B^{\text{even}}\amalg B^{\text{odd}}$. If we equip $\mathfrak{h}$ with a graded commutative pairing $\langle\cdot,\cdot\rangle$, we can apply the same construction as above to obtain the algebra

$$\hat{\mathfrak{h}}=(\mathfrak{h}_0[[t^{\pm1}]]\oplus\mathfrak{h}_1[[t^{\pm1}]])\oplus\mathbb{C}K \tag{2.0.7}$$

with Lie bracket

$$[at^m,bt^n]=\langle a,b\rangle\, m\delta_{m,-n}K, \qquad [K,\hat{\mathfrak{h}}]=0. \tag{2.0.8}$$

One can check that this is indeed a super Lie bracket (which essentially holds by the pairing being graded commutative and K being central), i.e. $\hat{\mathfrak{h}}$ is a Lie superalgebra.

Its Fock module is given by

$$
\begin{aligned}
H &\cong \operatorname{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-m}\right)\\
&\cong \bigotimes_{m>0}\operatorname{Sym}\left(\bigoplus_{a\in B^{\text{even}}}\mathbb{C}at^{-m}\right)\otimes\bigotimes_{m>0}\Lambda\left(\bigoplus_{a\in B^{\text{odd}}}\mathbb{C}at^{-m}\right).
\end{aligned}
$$

Now recall that while the symmetric power of a vector space yields the polynomial algebra on the basis of the vector space, the exterior power yields the algebra generated by the wedge products of the basic vectors. In the case of a 1-dimensional vector space generated by the element at^{-m}, we get

$$\Lambda(\mathbb{C}at^{-m})=\mathbb{C}\oplus\mathbb{C}at^{-m}$$

and thus the character contribution in this case is simply

$$1+q^m.$$

Multiplying throughout we obtain the character

$$\prod_{m=0}^{\infty}\frac{(1+q^m)^{\dim\mathfrak{h}^{\text{odd}}}}{(1-q^m)^{\dim\mathfrak{h}^{\text{even}}}}.$$

Let's introduce one last variant of this construction. This time we start from a graded vector space $\mathfrak{h}=\bigoplus_{m\in\mathbb{Z}}\mathfrak{h}_m$, which is automatically also a supervector space by the $\mathbb{Z}_2$-grading $\mathfrak{h}^{\text{even}}=\bigoplus_{i\in\mathbb{Z}}\mathfrak{h}_{2i}$ and $\mathfrak{h}^{\text{odd}}=\bigoplus_{i\in\mathbb{Z}}\mathfrak{h}_{2i+1}$. Suppose $\mathfrak{h}_i$ is finite-dimensional with basis B_i, and that $\mathfrak{h}$ is equipped with a graded commutative pairing $\langle\cdot,\cdot\rangle$. The definition of $\hat{\mathfrak{h}}$ as an algebra is given exactly as in Equation 2.0.7 with bracket given by Equation 2.0.8.

Now we obtain

$$H \cong \mathrm{Sym}\left(\bigoplus_{m>0}\bigoplus_{\substack{a\in B_i\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right)$$
$$\cong \bigotimes_{m>0}\mathrm{Sym}\left(\bigoplus_{\substack{a\in B_{2i}\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right)\otimes\bigotimes_{m>0}\Lambda\left(\bigoplus_{\substack{a\in B_{2i+1}\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right).$$

This gives the character

$$\sum_{\substack{\Delta\in\mathbb{Z}_{\geq}\\ i\in\mathbb{Z}}} \dim\mathfrak{h}_{i,\Delta}t^i q^\Delta = \prod_{m>0}\prod_j (1-(-1)^j t^j q^m)^{-(-1)^j\dim\mathfrak{h}_j}.$$

Finally fix $\mathfrak{h} = H_*(X)$, the singular homology ring of a complex projective surface X. The character of the corresponding Heisenberg algebra is

$$\prod_{m=1}^{\infty}\frac{(1+t^{2m-1}1^m)^{b_1(X)}(1+t^{2m+1}q^m)^{b_3(X)}}{(1-t^{2m-2}q^m)^{b_0(X)}(1-t^{2m}q^m)^{b_2(X)}(1-t^{2m+2}q^m)^{b_4(X)}} \tag{2.0.9}$$

It was proved by Göttsche that Equation 2.0.9 in fact equals

$$\sum_{n=0}^{\infty} q^n P_t(X^{[n]}),$$

where the *Poincaré polynomials* $P_t(Z)$ are defined by $P_t(Z) = \sum_{m\geq 0} b_m(Z)t^m$, $b_m(Z) = \mathrm{rank}H_m(Z)$ for any variety Z.

Since $\bigoplus_{n=0}^{\infty} H_*(X^{[n]})$ is the representation of the Heisenberg algebra we shall discuss in the main theorem, this shows that in fact it is a highest weight representation.

3. Hilbert scheme of points

In this section I define the Hilbert scheme of points of a projective complex surface. I give more details than what is really necessary for the remainder of this work. The essential property of the Hilbert scheme is that its points (over $\mathbb{C}$) are 0-dimensional closed subschemes of X of length n.

We start by constructing the Hilbert scheme following [Nak, Section 1.1]. Let X be a projective complex surface. Consider a contravariant functor

$$\mathcal{H}ilb_X : (\mathrm{Schemes})^{\mathrm{op}} \to (\mathrm{Sets})$$

which assigns to any scheme U the set of flat families of closed subschemes in X parametrized by U. Any such family is by definition a flat morphism

$$\begin{tikzcd} Z \arrow[d, "\pi"] & \subseteq X\times U \\ U. \end{tikzcd}$$

In particular, the fibres of such a morphism are subschemes of X. We define $\mathcal{H}ilb$ to map a morphism of schemes $U \to U'$ to a map of sets (of families over U and U') given by pullback.

The Hilbert polynomial at a fibre Z_u is defined by $P_u(m) = \chi(\mathcal{O}_{Z_u} \otimes \mathcal{O}_X(m))$. By flatness of π the Hilbert polynomial is constant along U [Har77, Part III, Theorem 9.9]. For a fixed polynomial P we define $\mathcal{H}ilb_X^P$ as the functor which maps U to the set of families over U with Hilbert polynomial P. A theorem by Grothendieck says $\mathcal{H}ilb_X^P$ is representable by a projective scheme Hilb_X^P. This means that there is a natural transformation $\mathrm{Hom}(-, \mathrm{Hilb}_X^P) \simeq \mathcal{H}ilb_X^P(-)$; that is, families over any scheme U are in bijection with morphisms $U \to \mathrm{Hilb}_X^P$.

As with any representable functor, we obtain a distinguished element associated to the identity morphism of the object representing the functor. In our case, this is a "universal" family $\mathcal{Z} \to \mathrm{Hilb}_X^P$ with the property that any other family $Z \to U$ is obtained as a pullback of $\mathcal{Z}$. More exactly, regarding an arbitrary family $Z \to U$ as a morphism $f \in \mathrm{Hom}(U, \mathrm{Hilb}_X^P)$, we see that

$$\begin{tikzcd}
\mathrm{id}_{\mathrm{Hilb}_X^P} \arrow[d, mapsto] & \mathrm{Hom}(\mathrm{Hilb}_X^P, \mathrm{Hilb}_X^P) \arrow[r, "\simeq"] \arrow[d, "f^*"'] & \mathcal{H}ilb_X^P(\mathrm{Hilb}_X^P) \arrow[d, "f^*"] & \mathcal{Z} \arrow[d, mapsto] \\
f & \mathrm{Hom}(U, \mathrm{Hilb}_X^P) \arrow[r, "\simeq"] & \mathcal{H}ilb_X^P(U) & Z = f^*\mathcal{Z}.
\end{tikzcd}$$

commutes since the Hilbert functor acts on morphisms by pullbacks of families. This shows that studying families of subschemes of X parametrized by any scheme U can be done by studying families over the Hilbert scheme.

In fact, we can formally view subschemes of X as points of Hilb_X^P as follows. Recall that, as with any other scheme, points h of Hilb_X^P correspond to morphisms $\mathrm{Spec}(\kappa(h)) \to \mathrm{Hilb}_X^P$, where $\kappa(h)$ is the residue field of the point (see e.g. Stacks Project 01J9). Then for every point h of the Hilbert scheme we have a unique subscheme of X defined as the fibre X_h of the corresponding morphism,

$$\begin{tikzcd}
X_h \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{Z} \arrow[d] & \subset X \times \mathrm{Hilb}_X^P \\
\mathrm{Spec}(\kappa(h)) \arrow[r] & \mathrm{Hilb}_X^P. &
\end{tikzcd}$$

As a rather technical observation that will not be addressed later, we note that to make sure that the fibres are $\mathbb{C}$-schemes, we must restricts our analysis to $\mathbb{C}$-points of the Hilbert scheme (i.e. when the residue field is $\mathbb{C}$).

Definition 3.1. Let P be a constant polynomial given by $P(m) = n$ for all $m \in \mathbb{Z}$. The *Hilbert scheme of n points in X* is $X^{[n]} := \mathrm{Hilb}_X^P$.

This condition forces the dimension of the fibres of any family over X to be zero. Indeed, the leading term of the Hilbert polynomial of a projective variety of degree d and dimension s is $(dm^s)/s!$, which can only be constant for $s = 0$ (see [HM98, Exercise 1.13]).

Further, any 0-dimensional subscheme of X is supported on finitely many points of X (this is consequence of X being Noetherian: we couldn't have infinitely many points since they would give an infinite chain of prime ideals). Finally, the *length* of any such subscheme, defined by $\dim \Gamma(\mathcal{O}_Z)$ must be $\sum_{x\in \mathrm{supp} Z} \dim \Gamma(\mathcal{O}_{Z_x}) = n$. (see Stacks Project 0AYT).

It is now easy to see that any subscheme of X consisting of n distinct points is itself a point of Hilb_X^P. Indeed, if $Z = \{x_1, \ldots, x_n\} \subset X$, its structure sheaf is the direct sum of skyscraper sheaves

$$\mathcal{O}_Z = \bigoplus_{i=1}^{n} (\text{skyscraper sheaf at } x_i)$$

so that $\mathcal{O}_Z \otimes \mathcal{O}_X(m) = \mathcal{O}_Z$.

However, not every point in $X^{[n]}$ is given by a set of n distinct points, nor by a set of points with multiplicites. Indeed, the Hilbert scheme distinguishes the "tangent directions in which points collide". The important thing to note here is that there are elements in the Hilbert scheme given by ideals which do not correspond to sets of points (even with allowed repetition).

The following construction is an important tool used to prove different statements related to the Hilbert scheme.

The *symmetric power* of X is

$$S^n X = \underbrace{X \times \ldots \times X}_{n \text{ copies}} / S_n$$

where the symmetric group S_n acts by permutating copies. Thus, a point of $S^n X$ is an S_n-orbit on the cartesian product $X \times \ldots \times X$, that is, an unordered n-tuple of points of X, with allowed repetition. In algebraic geometry these are called "algebraic cycles" and are denoted as formal sums $\sum_{x\in X} n_x[x]$ where $\sum_{x\in X} n_x = n$, with $n_x \in \mathbb{N}$ and $[x]$ are formal symbols.

By our discussion above on the length of 0-dimensional subschemes of X, we have a map

$$\begin{aligned} \pi : X^{[n]} &\longrightarrow S^n X \\ Z &\longmapsto \underbrace{\sum_{x\in \mathrm{supp} Z} \dim \Gamma(\mathcal{O}_{Z,x})[x]}_{\text{cycle of } Z} \end{aligned}$$

called the *Hilbert-Chow morphism.* In some cases this is an isomorphism, but not always, as the following theorem suggests:

Theorem 3.2 (Fogarty)**.** *Suppose X is nonsingular and $\dim X = 2$, then the following holds.*

(1) *$X^{[n]}$ is nonsingular and has dimension $2n$.*
(2) *The Hilbert-Chow morphism is a resolution of singularities.*

4. Borel-Moore homology

In this section we define Borel-Moore (BM) homology and formulate its basic properties. Most of this section is based on Video 9. For details on the construction see [Ful96].

Let M be a a smooth, connected, oriented, finite dimensional manifold over $\mathbb{R}$. While the following results hold for any topological space $X \subset M$ that is a proper deformation retract of a closed neighbourhood of a manifold, it's enough to think of $X \subset M$ as a (possible singular) variety over $\mathbb{C}$.

Recall that singular homology is defined as the homology of the chain complex of finite linear combinations of i-simplices in X. *Borel-Moore homology (BM)* $H_\bullet^{BM}(X)$ is the homology of the complex of locally finite linear combinations of singular simplices in X, where "locally finite" means that every point $x \in X$ is in a neighbourhood that intersects only finitely many simplices.

Perhaps the most striking difference between singular and BM homology is that the latter is not homotopy invariant. To see this recall that the homology groups of $\mathbb{R}^2$ are $H_0(\mathbb{R}^2) = \mathbb{C}$ and $H_i(\mathbb{R}^2) = 0$ for $i \neq 0$. However, $H_2^{BM}(\mathbb{R}^2) = \mathbb{C}$ since a tiling of triangles is a locally finite chain of 2 simplices, has no boundary (since the boundaries of all triangles cancel each other), and is clearly not the boundary of a 3-simplex; thus this tiling is a nonzero element of $H_2^{BM}(\mathbb{R}^2)$. Further, notice that $H_0^{BM}(\mathbb{R}^2) = 0$ since a ray of 1-simplices is a locally finite chain whose boundary is a point; thus a point is a boundary and thus has trivial homology class.

Here are three important properties of BM homology:

(1) If X is compact, $H_i^{BM}(X) \cong H_i(X)$.
(2) If X is not compact, $H_i^{BM}(X) \cong H_i(\hat{X}, \infty)$ where $\hat{X}$ is the one-point compactification of X and $H_i(\hat{X}, \infty)$ is the relative homology group.
(3) If the dimension of the manifold where X is embedded is m, there is a canonical isomorphism $H_i^{BM}(X) \cong H^{m-i}(M, M \setminus X)$. In other words, there is a perfect pairing

$$H_i^{BM}(X) \times H_{m-i}(M, M \setminus X) \to \mathbb{C}$$

given by "intersection" (where the quotes account for some transversal intersection condition that we would require before computing the intersection of two simplicial chains).

Consider the previous example of the tiling of $\mathbb{R}^2$ by triangles and put $X = M = \mathbb{R}^2$. Take any point as a 0-chain in singular homology and the triangle tiling as 2-chain in BM homology. Since the point lies in exactly one triangle (after suitable perturbation) their intersection is 1. Notice this wouldn't work with singular homology: we need an infinite 2-chain to make sure that the whole plane is tiled by triangles since otherwise we cannot guarantee that the point is contained in some triangle.

Now we formulate the key properties of BM homology needed to define the operators of Nakajima's representation of the Heisenberg algebra (these are also discussed in [Kam10, Section 4.2]):

(1) (Pushforward.) If $f : X \to Y$ is proper (i.e. inverse image of any compact set is compact) then there is a pushforward map $f_* : H_i^{BM}(X) \to H_i^{BM}(Y)$. This is functorial in proper maps, i.e. preserves compositions. Roughly, this works because, by properness, the inverse image of a point in Y will be mapped to a compact set of X, which can intersect only finitely many simplices of a given chain, ensuring local finiteness of the image of the chain under f.
(2) (Pullback.) If $f : X \to Y$ is a locally trivial fibration with fibre a manifold of dimension m (over $\mathbb{R}$), then $f^* : H_i^{BM}(Y) \to H_{i+m}^{BM}(X)$.

(3) (Intersection pairing.) If X and Y are both nicely embedded in the same manifold M,

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \to H_{i+j-m}^{BM}(X \cap Y).$$

Constructing this product can be done via the isomorphism with relative singular cohomology (item (3) in the previous list), and then using usual cap product.

(4) (Fundamental classes.) To any variety V we can associate a fundamental class $[V]$. More precisely,
- If V is a smooth irreducible variety of dimension d over $\mathbb{C}$, its *fundamental class* $[V]$ is the canonical generator of $H_{2d}^{BM}(V) \cong H^0(V) \cong \mathbb{C}$, where the first isomorphism is, again, given by the isomorphism of BM homology and relative singular cohomology taking $V = M$ since it is smooth.
- If V is an irreducible variety of dimension d over $\mathbb{C}$, let $[V] \in B_{2d}^{BM}(V)$ be the unique d such that $u^*[V] = [V_{\text{reg}}]$. That is, $[V]$ is the class corresponding to the pullback of u on the open set of regular points of V. The pullback in this degree is injective and thus this class is unique.
- For V not necessarily irreducible of dimension d, then $H_{2d}^{BM}(V)$ has basis $[V_1], \ldots, [V_k]$ where V_i are the d-dimensional irreducible components of V. In this case we define the fundamental class as the sum of the fundamental classes of those components, i.e. $[V] = [V_1] + \ldots + [V_k]$.

Notice that fundamental classes are defined in singular homology for compact manifolds. It is special feature of BM homology that we can define fundamental classes in the noncompact case.

5. Correspondences

In this section we introduce the main tool for constructing the operators in our representation of the Heisenberg algebra.

Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2. Let Z be a d-dimensional closed subvariety of $X_1 \times X_2$ such that the projections p_1 and p_2 are proper.

The *correspondence* associated to $Z \subset X_1 \times X_2$ is the map

$$\begin{aligned} c_{[Z]} : H_i^{BM}(X_2) &\longrightarrow H_{i+2d-2d_2}^{BM}(X_1) \\ c &\longmapsto p_{1,*}(p_2^* c \cap [Z]) \end{aligned} \tag{5.0.1}$$

where p_1 and p_2 are the projections in the following diagram:

$$\begin{tikzcd} & Z \subset X_1 \times X_2 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 & & X_2. \end{tikzcd}$$

That is, the correspondence is an operator given by pulling back a class by p_2, intersecting with the fundamental class of $[Z]$, and pushing forward by p_1.

Let us explain with more details why this map is well-defined. First we apply the pullback

$$H_i^{BM}(X) \xrightarrow{p_2^*} H_{i+2d_1}^{BM}(X_1 \times X_2),$$

which is valid by key property (2) of BM cohomology since the projection onto the second factor is a locally trivial (in fact trivial) fibration with fibres of real dimension $2d_1$. Also by key property (2) we see that the degree of the target is $i + 2d_1$.

Then we take the intersection pairing with the fundamental class of Z, which lies in the $2d$-th BM homology of Z. According to key property (3) this should be a map

$$H_{i+2d_1}^{BM}(X_1 \times X_2) \xrightarrow{\cap_{X_1\times X_2}[Z]} H_{i+d_1+2d-(2d_1+2d_2)}^{BM}(Z),$$

where the degree of the target is given by

$$\underbrace{i+2d_1}_{\substack{\text{we are here}\\\text{from previous step}}} + \underbrace{2d}_{\substack{\text{real dimension}\\\text{of } Z}} - \underbrace{(2d_1+2d_2)}_{\substack{\text{real dimension}\\\text{of ambient space}}},$$

which of course equals $i + 2d - 2d_2$. Finally take the pushforward map of p_1 using key property (1), since we assumed p_1 is proper.

Remark 5.1. (1) We can define correspondences more generally replacing the fundamental class of Z by any $\alpha \in H_{2d}^{BM}(Z)$. In fact, the operators in our Heisenberg represenation shall be of this more general type.
(2) If Z is the graph of a proper morphism $f : X_2 \to X_1$, then $c_{[Z]} = f_*$. This shows that correspondences generalize the pushforward in BM homology.
(3) Since X_1 and X_2 are smooth, we may use the isomorphism of BM and relative singular cohomology to identify a class in the tensor product with an endomorphism as we do in general for V, W vector spaces via the isomorphism $V \otimes W^* \cong \mathrm{Hom}(W, V)$, $v \otimes \varphi \mapsto (w \mapsto \varphi(w)v)$. This way, the fundamental class $[Z] \in \bigoplus_{i+j=\dim_{\mathbb{R}} Z} H_i(X_1) \otimes H_j^{BM}(X_2)$ immediately gives a map

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n\mp i]})$$

which, in fact, coincides with Equation 5.0.1. This is mentioned in

Next we show how to compose correspondences.

Suppose we are given two subvarieties $Z_{12} \subset X_1 \times X_2$ and $Z_{23} \subset X_2 \times X_3$ such that the projections to X_2 and X_3 are proper. We define the composition of the corresponding correspondences by

$$[Z_{12}] \circ [Z_{23}] := p_{13,*}(p_{12}^*[Z_{12}] \cap p_{23}^*[Z_{23}]) \tag{5.1.1}$$

according to the diagram

$$\begin{tikzcd}
 & X_1 \times X_2 \times X_3 \arrow[dl, "p_{12}"'] \arrow[dd, "p_{13}"] \arrow[dr, "p_{23}"] & \\
X_1 \times X_2 & & X_2 \times X_3 \\
 & X_1 \times X_3 &
\end{tikzcd}$$

which is well defined since, in fact, the projection $p_{13}(p_{12}^{-1}Z_{12} \cap p_{23}^{-1}Z_{23}) \to X_3$ is proper.

6. Main construction

We want to produce a representation π of the Heisenberg (super)algebra $\mathfrak{h}$ on $V = H_*(X)$. The key idea is to consider some subvarieties of direct products and use their fundamental classes to produce operators on the homology via correspondences.

Let $n \geq i > 0$. Define $P[i]$ as a subset of $\coprod_n X^{[n-i]} \times X^{[n]}$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1, \mathcal{J}_2, x) \in X^{[n-i]} \times X^{[n]} : \mathcal{J}_1 \supseteq \mathcal{J}_2, \operatorname{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\} \text{ for some } x \in X\}.$$

For $i < 0$, we swap $\mathcal{J}_1$ and $\mathcal{J}_2$ in this definition.

Notice that by the procedure in Section 5 the submanifold $P[i]$ already defines a correspondence by pullback, intersection with the fundamental class of $P[i]$ and then pushing forward. However, this is not quite what we will do: we will bring in two other classes. Let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and $i > 0$. Define

$$P_\alpha[i] = \varpi_*(\Pi_1^*\alpha \cap [P[i]])$$
$$P_\beta[i] = \varpi_*(\Pi_1^*\beta \cap [P[-i]])$$

according to the projections

$$\begin{tikzcd}
 & \coprod_n X^{[n-i]} \times X^{[n]} \times X \arrow[dl, "\Pi_1"'] \arrow[dr, "\varpi_i"] & \\
X & & \coprod_n X^{[n-i]} \times X^{[n]}.
\end{tikzcd}$$

I think the reason for this is that using just $P[i]$ as the correspondence could give non-proper projections, and the pullback may not be well-defined. Also, it's not clear why we have to choose α and β in different kinds of homology. Of course if we just assume that X is projective then BM homology coincides with usual homology.

In conclusion, the classes $P_\alpha[i]$ and $P_\beta[i]$ belong to $\prod_n H_*(X^{[n\mp i]}) \otimes H_*(X^{[n]})$. Here the $\mp$ means that if $i > 0$ we are substracting, and if $i < 0$ we are adding.

Now we use Poincaré duality (either in its BM version $H_i^{BM}(M) \cong H^{m-i}(M)$ or the usual $H_i(M) \cong H^{m-i}(M)$ one assuming M is compact) to identify a class in the tensor product with an endomorphism, as in $V \otimes W^* \cong \operatorname{Hom}(W, V)$, $v \otimes \varphi \mapsto (w \mapsto \varphi(w)v)$. This way $P_\alpha[i]$ and $P_\beta[-i]$ give maps

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n\mp i]}).$$

It's worth noting that in practice we will use the construction in Section 5, specifically in Remark 5.1(1) for the classes $P_\alpha[i]$ and $P_\beta[i]$. Namely, for $i > 0$ will use the projections p_1 and p_2 in

$$\begin{tikzcd} & X^{[n-i]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n-i]} & & X^{[n]}. \end{tikzcd}$$

and the formula

$$P_\alpha[i](c) = p_{1,*}(p_2^* c \cap \Pi_1^* \alpha). \tag{6.0.1}$$

The following proposition is used to compute the exact degrees in the domain and codomain of our correspondences:

Proposition 6.1. $\dim_{\mathbb{C}} P[i] \cap (X^{[n-i]} \times X^{[n]}) = 2n - i + 1.$

Proof. Suppose $i > 0$. Fix $\mathcal{J}_1 \in X^{[n-i]}$. By Theorem 3.2 we know that $\dim X^{[n-i]} = 2(n-i)$. To find the dimension of $P[i]$ we must add to this number the dimension of the variety obtained by varying $\mathcal{J}_2 \in X^{[n]}$ satisfying $\mathcal{J}_1/\mathcal{J}_2 = \{x\}$ for some $x \in X$. The generic case is when $x \in X \setminus \operatorname{supp}(\mathcal{O}_X/\mathcal{J}_1)$ (this may be accepted under the explanation that being $\mathcal{O}_X/\mathcal{J}_1$ the structure sheaf of a variety, it has less dimension than that of X, i.e. it is generic that x is not in this support; a formal proof is provided in [Nak, Lemma 8.31] during the proof of the main theorem). In fact, the set of such $\mathcal{J}_2$ may be identified with $\pi^{-1}(S^i_{(i)}X)$, where π is the Hilbert-Chow morphism and $S^i_{(i)} = \{[i]x \in S^i(X) : x \in X\}$. It is shown in [Nak, Lemma 6.10] that it must have complex dimension $i + 1$. □

Now we are almost ready to formulate the main theorem. First notice that this is [Nak, Theorem 8.13] and **not** [Nak95, Theorem 3.1]. Now let's clarify what are the main objects involved.

The pairing we use is the graded commutative pairing defined by Notice that this pairing is indeed graded commutative since the intersection pairing $\cap$ in BM homology is introduced via the usual cup product in relative cohomology (see key property (3) of BM homology); and of course it is valued in $\mathbb{C}$ since $H_*(\mathrm{pt}) = \mathbb{C}$.

Theorem 6.2 (Nakajima, Grojnowski)**.** *Suppose* $(deg\alpha)(deg\beta) \equiv 0(mod 2)$. *Then*

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i \delta_{i,-j} \langle \alpha, \beta \rangle \, Id. \tag{6.2.1}$$

where $\langle \alpha, \beta \rangle = p_*(\alpha \cap \beta)$ *for* $p : X \to pt$ *the unique morphism from* X *to the point* pt.

According to our constructions in Section 2, this means we have a representation of the Heisenberg algebra $\hat{\mathfrak{h}}$ associated to the graded vector space $H_*(X)$ on $\bigoplus_n H_*(X^{[n]})$.

Notice the factor $(-1)^{i-1}$ in Equation 6.2.1, which is not present in the Lie bracket equations of any of the variants of the Heisenberg algebra discussed earlier in this work. The reason for including this term in the main theorem is explained in [Nak, Theorem 9.15].

Having this factor causes no trouble at all: the Lie algebras defined using this bracket are isomorphic to the Lie algebras we introduced earlier. Indeed, we can define a Lie algebra isomorphism from, say, the most basic version of the Heisenberg algebra with bracket given by Equation 2.0.1, to the Lie algebra $\bigoplus_{n\in\mathbb{Z}} \mathbb{C}\tilde{h}_n \oplus \mathbb{C}K$ with bracket

$$[\tilde{h}_m, \tilde{h}_m] = (-1)^{m-1} m\delta_{m,-n}K, \qquad K \text{ central},$$

by putting

$$h_m \mapsto \begin{cases} (-1)^{m-1}\tilde{h}_m & m < 0 \\ \tilde{h}_m & m \geq 0. \end{cases}$$

7. Worked example

In this section we do a worked example following [Nak] and Video 10 to illustrate the main idea of the proof.

Fix $x \in X$ and let $\alpha = [X]$ be the BM class of X, and $\beta = [x]$ the class of the point x. Take a set of distinct n points $x_1, \dots, x_n \in X \setminus \{x\}$ none of which is x. Identify them with the corresponding ideal $\mathcal{J}$. We are going to interchangeably refer to those points as points and as the sheaf $\mathcal{J}$.

Let's calculate

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}].$$

Looking back at Equation 6.2.1, it's easy to see that we should obtain just Id.

Let's compute first $P_\beta[-1][\mathcal{J}]$. For this consider

$$\begin{tikzcd} & X^{[n+1]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n+1]} & & X^{[n]}. \end{tikzcd}$$

By definition, $P_{[x]}[-1]$ is given by: pull back $\mathcal{J}$ via p_2, intersect with the class $\varpi_*(\Pi^*\alpha \cap [P[-1]]) = P_{[x]}[-1]$ and pushforward down to $X^{[n+1]}$. Morally this means that we pushforward the cycle corresponding to

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \operatorname{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}\} \subset X^{[n]} \times X^{[n+1]} \times X.$$

The pushforward recovers the $X^{[n+1]}$ component, that is, the schemes $\mathcal{J}'$ in $X^{[n+1]}$ satisfying $\operatorname{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}$. Since x is not a point in $\mathcal{J}$, the only possible choice is that $\mathcal{J}' = \{x\} \cup \mathcal{J}$, that is,

$$P_{[x]}[-1][\mathcal{J}] = [\{x\} \cup \mathcal{J}].$$

On the other hand, the correspondence $P_{[X]}[1]$ starts with an ideal $\mathcal{J}' \in X^{[n+1]}$ intersects it with

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \operatorname{supp}(\mathcal{J}'/\mathcal{J}) = \{y\}, \text{ for some } y \in X\}$$

and projects to $X^{[n]}$. We obtain $\sum_{y\in\mathcal{J}'}[\mathcal{J}' - \{y\}]$. Then it becomes clear that

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$

The following figure is taken directly from [Nak]):

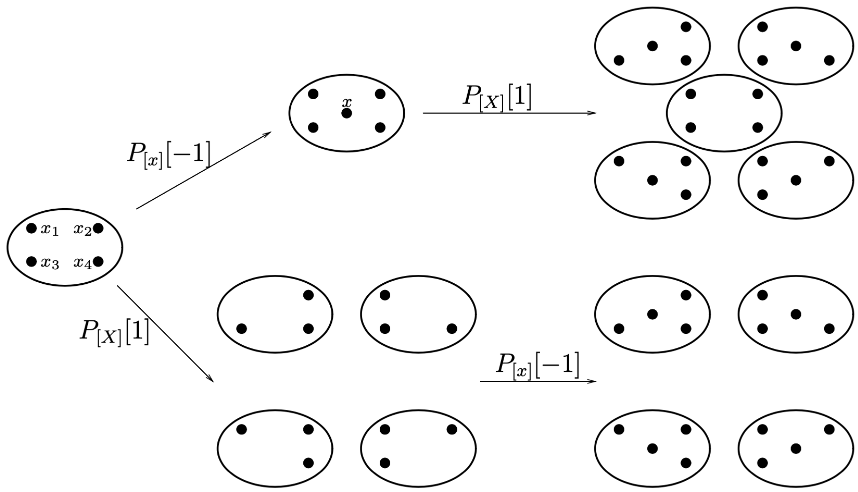

FIGURE 8.1. The proof of $\left[P_{[X]}[1], P_{[x]}[-1]\right][\mathcal{J}_1] = [\mathcal{J}_1]$

The idea is very simple: it is not the same to add the point x and then remove any point, than removing any point and then adding the point x. Indeed, in the latter case we are obliged to have x, while in the former we may or may not. Thus the operators do not commute.

The choice of α and β is special in this example. The annihilating operator being associated to the class of the whole space, $[X]$, allows us to remove any point $y \in X$. If we chose another class β and the support of the quotient $\mathcal{J}'/\mathcal{J}$ contained only points that are not in β, the operator would kill the class. This means that if we iterate an annihilating operator enough times we arrive at the zero class.

8. Proof of the main theorem

We split the proof in two different cases:

(1) $i, j > 0$ or $i, j < 0$, and
(2) $i > 0$, $j < 0$ or $i < 0$, $j > 0$.

Only in case (2) can the commutator be nonzero.

Case (1). Suppose $i, j > 0$. Recall our general definition for composition of correspondences, Equation 5.1.1. In view of the particular correspondences we are using, i.e. Equation 6.0.1, we see that the composition $P_\alpha[i]P_\beta[j]$ is given by

$$P_\alpha[i]P_\beta[j] = p_{13,*}(p_{12}^*[P[i]] \cap \Pi_1^*\alpha \cap p_{23}^*[P[j]] \cap \Pi_2^*\beta). \tag{8.0.1}$$

Consider the set

$$\begin{aligned} P[i,j] &:= p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]}) \\ &= \left\{ (\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3) : \mathcal{J}_1 \supset \mathcal{J}_2 \supset \mathcal{J}_3, {}^{\mathrm{supp}(\mathcal{J}_1/\mathcal{J}_2)=\{x\} \text{ for some } x\in X}_{\mathrm{supp}(\mathcal{J}_2/\mathcal{J}_3)=\{y\} \text{ for some } y\in X} \right\} \end{aligned} \tag{8.0.2}$$

which corresponds to the diagram

$$\begin{tikzcd} & X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]} \arrow[dl, "p_{12}"'] \arrow[dr, "p_{23}"] \arrow[d, "p_{13}"] & \\ P[j] \subset X^{[n-j]} \times X^{[n]} & X^{[n-i-j]} \times X^{[n]} & X^{[n-i-j]} \times X^{[n-j]} \supset P[i] \end{tikzcd}$$

This is to be read as follows. By definition, the correspondence $P[j]$ maps the homology of $X^{[n]}$ to the homology of $X^{[n-j]}$. In order to compose this with $P[i]$ we must regard $P[i]$ as a map from the homology of $X^{[n-j]}$ to the homology of $X^{[n-j-i]}$, which is perfectly valid since our correspondences are defined for all n. Observe that the middle factor $X^{[n-j]}$ in the triple product will become $X^{[n-i]}$ when we compute the composition in the opposite order.

To recover the correspondence as in Equation 8.0.1 from the set $P[i,j]$ it's natural to consider its algebraic closure and to it associate a fundamental class. We will do this in the following special way. Observe that $P[i,j]$ may be decomposed as disjoint union of the set $\{x \neq y\}$ of "generic" cases in which x and y in Equation 8.0.2 are different, and the set $\{x = y\}$ in which they coincide. Let $\overline{P}[i,j] := \overline{P[i,j] \cap \{x \neq y\}}$. Then

$$p_{12}^*[P[i]] \cap p_{23}^*[P[j]] = [\overline{P}[i,j]] + \iota_* R$$

where $R \in H_*^{BM}(P[i,j] \cap \{x = y\})$ corresponds to the $\{x = y\}$ part of $P[i,j]$ and ι is the inclusion.

To arrive at Equation 8.0.1 the next step is to intersect with the classes $\Pi_1^*\alpha$ and $\Pi_2^*\beta$. Observe that these commute with the classes $p_{12}^*[P[i]]$ and $p_{23}^*[P[j]]$ since they correspond to disjoint components of the product $X^{[n-j]} \times X^{[n]} \times X$. Then it only remains to pushforward by p_{13}.

We can carry out the same construction for the composition $P_\beta[j]P_\alpha[i]$, obtaining

$$P_\alpha[j]P_\beta[i] = p_{13,*}(p_{12}^*[P[j]] \cap \Pi_2^*\alpha \cap p_{23}^*[P[i]] \cap \Pi_1^*\beta). \tag{8.0.3}$$

and

$$p_{12}^*[P[j]] \cap p_{23}^*[P[i]] = [\overline{P}[j,i]] + \iota'_* R'. \tag{8.0.4}$$

Observe that Equations 8.0.1 and 8.0.3 differ in the classes $\Pi_1^*\alpha$ against $\Pi_2^*\alpha$, and correspondingly with β. The following lemma is all we need to obtain the result:

Lemma 8.1. (1) *There exists an isomorphism*

$$P[i,j] \cap \{x \neq y\} \xrightarrow{\cong} P[j,i] \cap \{x \neq y\}, \tag{8.1.1}$$

which exchanges Π_1 *and* Π_2.

(2) *We have*

$$p_{13,*}(\Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R) = 0, \qquad p_{13,*}(\Pi_1\beta \cap \Pi_2^*\alpha \cap \iota_* R') = 0. \tag{8.1.2}$$

Indeed, this lemma is all we need since

$$\begin{aligned} P_\alpha[i]P_\beta[j] &= p_{13,*}(\overline{P}[i,j]\cap \Pi_1^*\alpha\cap\Pi_2^*\beta) \\ &= (-1)^{\deg\alpha\deg\beta}p_{13,*}(\overline{P}[j,i]\cap \Pi_1^*\beta\cap\Pi_2^*\alpha) \\ &= P_\beta[j]P_\alpha[i]. \end{aligned}$$

Proof of Lemma 8.1. (1) First notice that when we say the isomorphism, say φ, "exchanges Π_1 and Π_2" we mean that $\Pi_1|_{X\times X} = \Pi_2|_{X\times X}\circ\varphi$ according to the diagram

$$\begin{tikzcd}
 & & X \\
P[i,j]\cap\{x\neq y\}\subset \arrow[ddd, "\varphi"'] & X^{[n-i-j]}\times X^{[n-j]}\times X^{[n]}\times X\times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X \\
 & & X \\
P[j,i]\cap\{x\neq y\}\subset & X^{[n-i-j]}\times X^{[n-i]}\times X^{[n]}\times X\times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X.
\end{tikzcd}$$

Now, the point of the isomorphism φ is that it must map an ideal in $X^{[n-j]}$ to an ideal in $X^{[n-i]}$. More precisely, we map $(\mathcal{J}_1,\mathcal{J}_2,\mathcal{J}_3)\in P[i,j]\cap\{x\neq y\}$ to $(\mathcal{J}_1,\mathcal{J}_2',\mathcal{J}_3)\in P[j,i]\cap\{x\neq y\}$. The defining relation is

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_2/\mathcal{J}_3, \qquad \mathcal{J}_2'/\mathcal{J}_3 = \mathcal{J}_1/\mathcal{J}_2. \tag{8.1.3}$$

(No further explanation other than a figure is provided in [Nak].)

The setting is the following. We have the ideals with contentions $\mathcal{J}_1\supset\mathcal{J}_2,\mathcal{J}_2'\supset\mathcal{J}_3$. Passing to the supports, the inclusions are reversed, i.e the support of $\mathcal{J}_3$ contains the rest, and the support of $\mathcal{J}_1$ is contained in the rest.

To prove that this function is well-defined we need to show that for every $\mathcal{J}_2$ there is a unique $\mathcal{J}_2'$ satisfying Equation 8.1.3. Notice that by definition, $\operatorname{supp}\mathcal{J}_2/\mathcal{J}_3=\{x\}$ for some $x\in X$ so that, as a scheme, $\{x\}$ has multiplicity j. We must define $\mathcal{J}_2'=\mathcal{J}_1\cup\{x\}$ where x has multiplicity j. (Caveat: how can we make sure that indeed there is only one scheme of length j supported in the point x? Recall that elements in the Hilbert scheme should not be considered barely as sets of points with multiplicities! Actually, in Proposition 6.1 we used the fact that the space of possible choices coincides $\pi^{-1}(S^j_{(j)})$ and has dimension $j-1$.)

Defining $\mathcal{J}_2'$ this way implies that

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_1/(\mathcal{J}_1\cup\{x\}) = \{x\} = \mathcal{J}_2/\mathcal{J}_3.$$

Now, we also must have $\mathcal{J}_1/\mathcal{J}_2 = \{y\}$ for some $y \in X$ with multiplicity i. This implies that $\mathcal{J}_1/\mathcal{J}_3 = \{x\} \cup \{y\}$ with their multiplicities. Then

$$\mathcal{J}_2'/\mathcal{J}_3 = (\mathcal{J}_1 \cup \{x\})/\mathcal{J}_3 = \{y\} = \mathcal{J}_1/\mathcal{J}_2$$

as required. An inverse function defined similarly shows this is a bijection.

Here I am supposing that "isomorphism" means bijection since $P[i,j]$ is considered as a set-theoretical intersection.

(2) Consider the commutative diagram

$$\begin{tikzcd} P[i,j] \cap \{x=y\} \arrow[r, "\Pi'"] \arrow[d, "\iota"'] & X \arrow[d, "\Delta"] \\ P[i,j] \arrow[r, "\Pi_1 \times \Pi_2"'] & X \times X \end{tikzcd}$$

where Δ is the diagonal embedding and Π' is any of the projections Π_1 or Π_2 (which coincide on $P[i,j] \cap \{x=y\}$). Then

$$\begin{aligned} \Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R &= \iota_*(\iota^*(\Pi_1^*\alpha \cap \Pi_2^*\beta) \cap R) \\ &= \iota_*(\iota^*(\Pi_1 \times \Pi_2)^*(\alpha \times \beta) \cap R) \\ &= \iota_*({\Pi'}^*\Delta^*(\alpha \times \beta) \cap R) \\ &= \iota_*({\Pi'}^*(\alpha \cap \beta) \cap R). \end{aligned} \tag{8.1.4}$$

We also have the commutative diagram

$$\begin{tikzcd} p_{13} \circ \iota : P[i,j] \cap \{x=y\} \arrow[r, "\Pi'"] \arrow[d, "p_{13\circ\iota}"'] & X \arrow[d, equal] \\ P[i+j] = \{(\mathcal{J}_1, \mathcal{J}_3) | \mathcal{J}_1 \supset \mathcal{J}_3, \operatorname{supp}(\mathcal{J}_1/\mathcal{J}_3) = \{x\}\} \arrow[r, "\Pi''"'] & X, \end{tikzcd}$$

where Π'' is the usual projection to X from the set $P[i+j]$.

By Equation 8.1.4 and the commutativity of the second diagram, we can express the class of interest as intersection of two classes: one pulled back **directly from** X, and the other one corresponding to R. That is, we have

$$p_{13,*}\iota_*({\Pi'}^*(\alpha \cap \beta) \cap R) = {\Pi''}^*(\alpha \cap \beta) \cap p_{13,*}\iota_*(R).$$

Now, by Proposition 6.1, we know that

$$\dim_{\mathbb{C}} P[i+j] = 2n - i - j + 1.$$

On the other hand, the "expected dimension" is $2n-i-j+2$. This obliges $\iota_* R$ to vanish.

$\square$

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